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Current Evidence for the Use of Jugular Vein Compression Collars in Sport: A Systematic Review

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Abstract

Sport-related concussions are a common type of brain injury, and the best treatment is prevention. Recently, external jugular vein compression collars have been worn by National Football League players, but the current evidence is limited. To the best of the authors' knowledge, this is the first comprehensive, up-to-date systematic review addressing the use of jugular vein compression collars for decreasing concussion incidence in high-impact sports and activities. Overall, when compared to control, the results indicated that jugular vein compression collars resulted in significantly reduced white matter alteration, improved short-term neurocognitive outcomes, increased internal jugular vein cross-sectional area, decreased internal carotid artery cross-sectional area, and moderating effects on peak pulse pressure. There was no significant difference in concussion incidence rate between groups. While promising, these findings warrant future research to assess jugular vein compression collar's role in concussion prevention, brain injury, and long-term neurocognitive outcomes. The authors are hopeful for the role jugular vein compression collars will play in the future.

Introduction

According to the most recent Amsterdam Consensus Statement on Concussion in Sport, a sport-related concussion is defined as a traumatic brain injury due to external forces to the head, neck, or body that result in force being transmitted to the brain occurring in activities related to sport or exercise (1). Although no abnormalities are seen on computed tomography or magnetic resonance imaging, neurotransmitter, metabolic, blood flow, and inflammatory changes occur (1). These injuries are of high concern, especially in young athletes. In the United States of America, 3.8 million sport-related concussions are annually reported in patients aged 0 to 18 years (2–5). In high school athletes, estimates are around 400,000 (2). High-impact sports (*e.g.*, American football, rugby, hockey, lacrosse, soccer), motor vehicle collisions, and falls have been established to be activities with higher incidence of concussions (3). Sports with the highest risk of concussion include youth hockey, college football, high school football, lacrosse, basketball, wrestling, soccer, softball, and volleyball (2). In high school athletes, concussions encompass 8.9% of athletic injuries but only 5.8% of collegiate sport injuries (4).

Concussions are the most prevalent type of sport-related traumatic brain injury (4). The incidence of sport-related concussions ranges from 4.47 per 10,000 athletic exposures (5). Although 7 to 14 d is the commonly reported time for recovery after injury, most sport-related concussions typically take 3 to 4 wk for complete symptom resolution (6), with some lasting longer or even developing chronic symptoms. Slower recovery from concussion has been associated with the severity of acute and subacute symptoms. In particular, headache and depression development postinjury are risk factors for concussion symptoms lasting beyond 4 wk (7). Additionally, individuals with mental health illness are at higher risk of ongoing symptoms (7). Although strict rest until resolution of concussion-related symptoms was previously recommended, current guidelines suggest only relative rest for the first 48 h after injury. Thereafter, athletes can increase physical activity based on the degree of symptom exacerbation experienced in their previous training session (1). The progression for an athlete typically involves 24 or more hours at the following graded steps: symptom-limited activity, aerobic exercise, sport-specific exercise, noncontact training drills, full contact practice, and return to sport (1).

Concussion prevention and proper treatment are vital to the health of athletes as previously diagnosed concussions and sport participation are risk factors for future concussions

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(1,8). Memory, executive function, and psychomotor function are cognitive domains negatively affected by history of concussion and head impact (9). When compared to controls, concussed athletes experience significant decreases in memory, executive function, self-reported cognitive function, language, and psychomotor function (10). Mental health illness also has been associated as a sequela with the aforementioned brain injury (11). Although there is ongoing controversy over the specific definition, when patients experience three or more symptoms for greater than 3 wk after concussion, namely, headache, affective abnormalities, dizziness, fatigue, insomnia, irritability, and memory deficits, they meet the World Health Organization's International Classification of Disease (ICD-10) definition of postconcussion syndrome (PCS) (12,13). Even though PCS is an unwanted ramification, the most feared potential long-term sequelae from repeated concussion is development of chronic traumatic encephalopathy (CTE). This neurodegenerative disease can only be officially diagnosed postmortem, is related to hyperphosphorylated tau protein and intracellular neurofibrillary tangles, and is proposed to be secondary to repetitive head injury, with both subconcussive and concussive impact being contributory (14,15). It bears mentioning that CTE also has been seen on autopsies of patients without known head trauma. CTE can cause multiple ramifications, which include behavioral fluctuation, cognitive dysfunction, and altered motor function, and when compiled together, it may even lead to death (14). Even amidst the aforementioned negative outcomes secondary to repeated concussion, the treatment remains uninvasive. Physical and cognitive rest with gradual return to activity has been well established as the primary management (12,16). Interestingly, excessively prolonged rest may negatively affect outcomes (17). Since the treatment course is primarily supportive, with controlled physical and cognitive loads, the best treatment for concussions may be prevention.

The mainstay preventative strategies of sport-related concussions are sport-specific helmets, mouthguards, policy/rule changes, training, and education (18). Of note, helmets in sports such as skiing, snowboarding, motorbiking, and bicycling have been found to be protective of catastrophic head injuries but do not prevent concussions (17). Additionally, neck strengthening is conditionally recommended in adult athletes, especially in women's soccer (19). Even with advanced protective equipment and a better understanding of brain injury, the gap in the prevention of sport-related concussions remains large. With potential long-term sequelae secondary to recurrent concussions, it is vital to protect those at risk. Recently, there has been research focused on mitigating brain injury via decreasing sloshing of the brain in the intracranial space in rats (20). This was achieved by maintaining higher venous pressures secondary to mechanical venous outflow obstruction. Researchers found there may be an association with the compliance of the intracranial fluid space and likelihood and severity of concussion (20). As indicated by immunohistological analysis, there was more than an 80% decrease in the amyloid protein precursor proteins found in the neuronal axons of rats with internal jugular vein compression (20). With this in mind, Smith et al. performed an epidemiological group study analyzing the concussion incidence at higher altitudes to assess their speculation, of which they coined the "Tighter Fit" theory (21). Researchers found that high altitude-related physiologic responses may be associated with decreased sport-related concussion rates in high-impact sports (21). Since findings of similar nature, researchers have immersed themselves in this idea, which played a part in the development of the external jugular vein compression collar (JVCC) for concussion reduction.

Recently, National Football League (NFL) players have been noted to be wearing JVCCs, which has sparked interest in both the public and scientific communities. To the best of the authors' knowledge, there is no currently available comprehensive systematic review for the use of JVCCs for concussion reduction in collision sports. Our objective is to provide an all-encompassing systematic review that assesses the benefits of utilizing JVCC in head impact-prone sports and activities for mitigating brain injury. The authors hypothesize that JVCCs are an effective way to reduce impact-related concussion in sports and activities. With the combination of sport-related concussions and the potential of long-term sequelae from these injuries, the need for optimal prevention strategies for concussions is at an all-time high. This manuscript provides an up-to-date review on the current evidence of JVCCs.

Methods

Research Question

Is the utilization of external jugular vein compression collars in high-impact sports and activities helpful in mitigating concussions?

Hypothesis

External jugular vein compression collars are an effective way to reduce impact-related concussion in contact sports and activities.

Inclusion Criteria

Population: healthy adults aged 15 to 30 with (experimental) and without (control) a concussion; factor or model: external jugular vein compression collar; outcome: clinically diagnosed concussion; study design: observational studies (groups, cross-sectional, case control, or case series). Publication date: 2000 to 2024; language: studies available in only English.

Exclusion Criteria

Population: age <15 or >30 years old; factor or model: absent use of external jugular vein compression collar; outcome: general TBI or nonspecified concussion; study design: overviews, narrative literature reviews, comments, and editorials.

Search Strategy

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) criteria was used for article selection in this systematic review (22). The following designed search strategy was used. The search strategy focused on published studies up until June 2024. A multiple-database search (PubMed, ScienceDirect, and Google Scholar) was done using the following search syntax: "compression collar concussion," "compression vein concussion," "jugular concussion," "collar concussion," "compression jugular concussion," "Q collar concussion," or "neck compression." This systematic review considered observational studies available in English published between 2000 and 2024. Additionally, systematic reviews that met the inclusion criteria were considered. Data were extracted from papers included in the systematic review by two independent reviewers (B.D. and F.M.).

Data Sources

To account for studies not listed during the original database search, manual searches for relevant articles were done.

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Additionally, manual searching through the reference of included articles was done.

Study Selection

Articles were included and excluded according to the inclusion or exclusion criteria, respectively. Duplicates were recognized and were not included more than once. There were not any disagreements among reviewers (B.D. and F.M.).

Data Extraction

Data related to study design, participants, methodology, sport/activity, intervention (control vs experimental), results, funding source, and declaration of interests were manually extracted from each included study.

Quality Assessment

Using Microsoft Excel (Microsoft Corp., Redmond, WA), the Cochrane Risk of Bias Assessment Tool version 2 was run by two independent reviewers (B.D. and F.M.) to define the risk of bias within each included study (23). There were no disagreements among reviewers (B.D. and F.M.).

Results

Literature Search

The search strategy yielded 1971 results. Of those, 114 were duplicate studies, and 1951 were removed for non-relevance (Fig. 1). A total of 20 records were screened and sought for retrieval (Fig. 1). Of the 20 articles screened, a total of 5 were excluded (3 editorials, 2 animal studies). A total of 15 studies were included in the review (Fig. 1).

Characteristics of Studies Included

The characteristics of the 15 induced studies can be found in Table 1 (24–38). All selected studies were randomized controlled trials, and 11 studies were identified as prospective controlled trials. Studies were on high school male football seasons, male and female soccer seasons, male hockey seasons, and special weapons and tactics (SWAT) teams. Studies also were conducted on physiological and anatomical responses to JVCC (24,36). Participants were followed for one high school football season (25,28,31,35), one female high school soccer season (29,30), two consecutive high school football seasons (34), and a combination of football and soccer players (38). All participants were randomized to JVCC or control groups.

Risk of Bias Assessment

The risk of bias was assessed using the Cochrane Risk of Bias Assessment Tool version 2 (23). Overall, the bias was found to be low risk with some concern for the intention-to-treat (Fig. 2A) and per-protocol (Fig. 2B) studies. Of the 15 included studies, study I was the only one to possess high risk of bias (Fig. 2A). It should be noted that Q30 Sports Science, LLC, funded the majority of the high school studies, as described in Table 1. Dr. David Smith is the founder of the Q-Collar and is a published author on 7 of the 15 studies (Table 1).

Longitudinal Sports Seasons and Neuroimaging Results

Head impact exposure was found to be insignificant between the two interventions (25,27–32,34), except >50 g-force difference of head impact exposure in the JVCC group ($P = 0.024$) (Table 2) in study I (32). White matter (WM) alterations were found to be greater in the control group compared to the JVCC group during mid-season (9) and after one season (26,28,30,32). Study D showed decreased mean diffusivity, axial diffusivity, and radial diffusivity (MD, AD, and RD) from preseason year 1 (PrY1) to postseason year 1 (PoY1), PrY1 to PrY2, and from PoY1 to PrY2 ($P < 0.05$) (27). Study E reported the control group had significantly reduced PrY1 to PoY1 MD ($P = 0.011$), AD ($P = 0.021$), RD ($P = 0.018$), and significantly increased in FA ($P = 0.0046$) (28). The control group had significantly increased resting-state functional

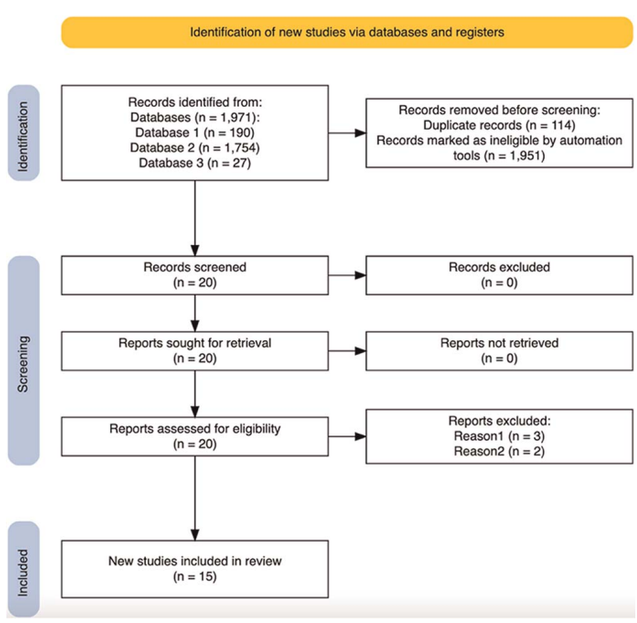

Figure 1: PRISMA flow diagram.

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Table 1.
Characteristics of studies included.

Study	Design	Group Size (n)	Males, Females	Control	Experimental	Sport or Activity	Intervention	Funding Source	Declaration of Interest
A (24)	PL	19	13, 6	None	19	None	JVCC	None	None
B (25)	PL	284	284, 0	144	140	Football	JVCC	Q30 Sports Science, LLC	GDM has consulted with Q30 Sports Science, LLC, but has no financial interest
C (26)	PL	46	32, 14	24	22	Football and soccer	JVCC	Q30 Sports Science, LLC	GDM has consulted with Q30 Sports Science, LLC, but has no financial interest
D (27)	PL	PrY1 42 PoY1 42 PrY2 23 PoY2 16	PrY1 42:0 PoY1 42:0 PrY2 23:0 PoY2 16:0	PrY1 21 PoY1 21 PrY2 10 PoY2 I	PrY1 21 PoY1 21 PrY2 13 PoY2 9	Football	JVCC	The Heidt Family Foundation, Robert S. Heidt Sr. Wellington Foundation, and Q30 Sports Science, LLC	DS is the inventor of the Q-Collar and has a financial interest in study results; JB is a stockholder in Q30 Sports Science, LLC
E (28)	PL	284	284, 0	142	142	Football	JVCC	Q30 Sports Science, LLC	GDM has consulted with Q30 Sports Science, LLC, but has no financial interest
F (29)	PL	128	0, 128	56	72	Soccer	JVCC	Q30 Sports Science, LLC	GDM has consulted with Q30 Sports Science, LLC, but has no financial interest
G (30)	PL	46	0, 46	22	24	Soccer	JVCC	The Heidt Family Foundation, Robert S. Heidt Sr. Wellington Foundation, and Q30 Sports Science, LLC	DS is the inventor of the Q-Collar and has a financial interest in study results
H (31)	PL	52	52, 0	27	25	Football	JVCC	The Heidt Family Foundation, Robert S. Heidt Sr. Wellington Foundation, and Q30 Sports Sciences LLC	DS is the inventor of the Q-Collar and has a financial interest in study results
I (32)	PL	62	62, 0	30	32	Football	JVCC	Q30 Sports Science, LLC	DS is the inventor of the Q-Collar and has a financial interest in study results
J (33)	RCT	BEG: 23 fMRI-TWM: 18 fMRI-RS: 20	23, 0	11	12	SWAT	JVCC	Q30 Sports Science, LLC	DS is the inventor of the Q-Collar and has a financial interest in study results
K (34)	PL	48	0, 48	21	27	Soccer	JVCC	The Heidt Family Foundation, Robert S. Heidt Sr. Wellington Foundation, Q30 Sports Science, LLC	DS is the inventor of the Q-Collar and has a financial interest in study results; GDM has consulted with Q30 Sports Science, LLC, but has no financial interest
L (35)	PL	15	15, 0	PrEEG 8 PrMRI 7 PoEEG 7 PoMRI 7	PrEEG 7 PrMRI 7 PoEEG 3 PoMRI 2	Hockey	JVCC	The Heidt Family Foundation and Q30 Sports Science, LLC	DS is the inventor of the Q-Collar and has a financial interest in study results
M (36)	RCT	16	8:8	8	8	Healthy volunteers	JVCC	York University Faculty of Health, Canadian Foundation for Innovation, and Ontario Research	Q-Collars were supplied by Q30 Sports Science, LLC
N (37)	RCT	284	284, 0	142	142	SWAT	JVCC	Q30 Sports Science, LLC	None

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Study	Design	Group Size (n)	Males, Females	Control	Experimental	Sport or Activity	Intervention	Funding Source	Declaration of Interest
O (38)	RCT	21	21, 0	10	11	SWAT	JVCC	Q30 Sports Science, LLC	GDM has consulted with Q30 Sports Science, LLC, but has no financial interest

PL, prospective longitudinal; RCT, randomized controlled trial; PrY1, preseason year 1; PoY1, postseason year 1; PrY2, preseason year 2; PoY2, postseason year 2; PrEEG, preseason electroencephalogram; PoEEG, postseason electroencephalogram; PrMRI, preseason magnetic resonance imaging; BEG, beginning of season; fMRI-TWM, functional magnetic resonance imaging total working memory; fMRI-RS, functional magnetic resonance imaging resting state; PoMRI, postseason magnetic resonance imaging; DS, Dr. David Smith; GDM, Dr. Gregory D. Myer; JB, Dr. Julian Bales.

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magnetic resonance imaging (rs-fMRI) global clustering coefficients ($P < 0.032$), modularity ($P < 0.05$), and significantly increased DTI modularity ($P = 0.042$) than the JVCC group (29). Study G reported a significant decrease in MD, AD, and RD in the control group from preseason to PoY1 ($P < 0.05$) (30). Study I found a significant PrY1 and PoY1 change in WM integrity and DTI changes for the control group ($P < 0.05$) (32). Study L found a significant increase in RD and MD from pre- to mid-season in the control group ($P < 0.05$) (35). Additionally, study L found an increased change in BNA scores in the control group compared to the JVCC group ($P < 0.006$) (Table 2) (35). Study H found the control group had an increased fMRI BOLD response that was associated with the total number of impacts and g-force ($P < 0.05$) (31). Study K found significantly higher PrY1 to PoY1 brain activation compared to the JVCC group ($P < 0.05$) (34).

Longitudinal Sports Seasons and Cognitive Scores

The JVCC group was found to have higher ANT scores than the control group ($P = 0.026$) (Table 2) (25). JVCC participants were found to have reduced KD exam time scores ($P = 0.003$) and increased conflict scores ($P = 0.01$) (26).

Longitudinal Sports Seasons and Concussion Incidence

The 2-year longitudinal high school football team analysis did not report a change in concussion incidence rate between JVCC and control interventions. However, WM alterations were greater in control versus JVCC ($P < 0.05$) (Table 2) (26).

JVCC and Physiological Responses

The internal jugular vein increased from 0.10 to 0.57 cm^2 ($P < 0.001$) with the use of a JVCC (Table 2) (24). The JVCC group was found to have significantly increased the jugular vein cross-section area in men and women at the end of inhalation ($P = 0.003$) and reduced the carotid artery cross-sectional area at the end of exhalation ($P = 0.022$) (36). The JVCC group reported a significant moderating effect on increased pulse pressure compared to the control group ($P = 0.048$) (Table 2) (38).

SWAT Training and Neuroimaging Results

No significance in blast exposure was reported between the JVCC and control groups ($P > 0.05$) (Table 2) (33,37,38). Control participants had an increased postblast training fMRI 2-back > 0-back activation compared to preblast training ($P < 0.05$) (33). A correlation was reported in the control group

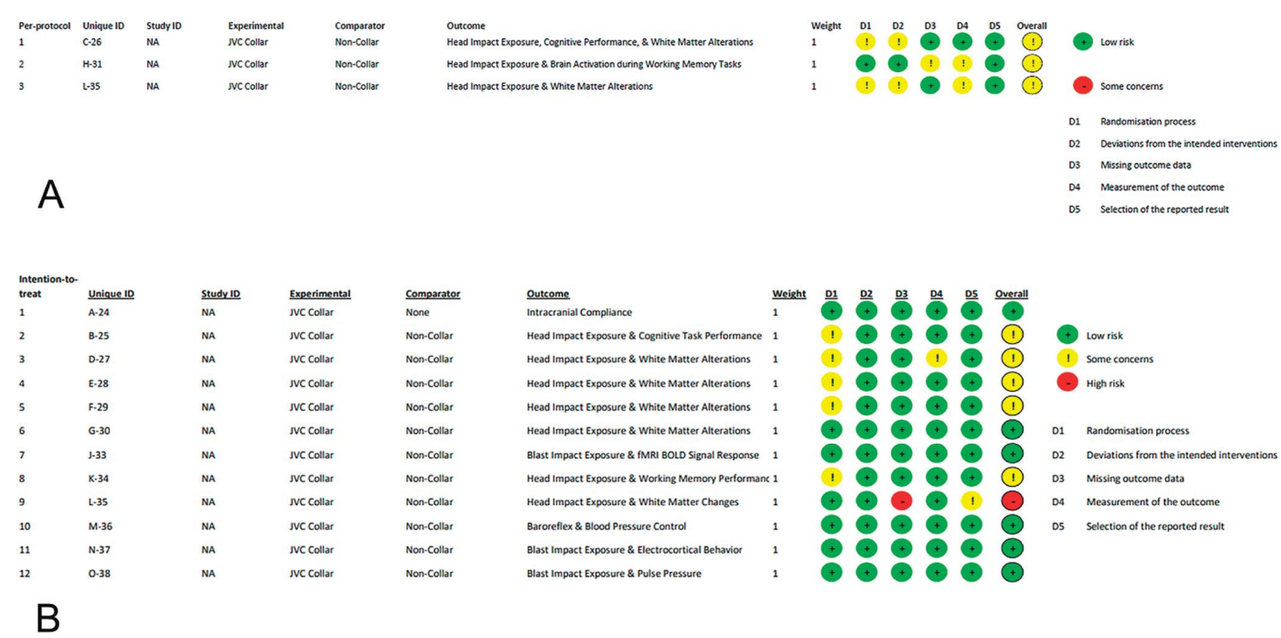

Figure 2: (A) Report of intention-to-treat bias using the Cochrane Risk of Bias Measurement Tool 2.0 (Cochrane et al., 2011). (B) Report of per-protocol bias using the Cochrane Risk of Bias Measurement Tool 2.0 (Cochrane et al., 2011). D1: randomization process; D2: deviations from the intended interventions; D3: missing outcome data; D4: measurement of the outcome; D5: selection of the reported result; + low risk; ! some concerns; −: high risk.

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Table 2.
Outcome and results of studies included.

Study	Outcome	Results
A (24)	Intracranial response to IJV compression	● IJV cross-sectional area: ○ Before: 0.10 cm (2) ($P < 0.001$) ○ After: 0.57 cm (2) ($P < 0.001$)
B (25)	Behavior and cognitive function	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● No significance with all cognitive exams, expect the JVCC group had higher ANT scores ($P = 0.026$)
C (26)	Neuroimaging, behavioral, and cognitive scores	● Concussion incidence rate comparable between JVCC versus control ● WM alterations were greater in control versus JVCC ($P < 0.05$) ● JVCC had significantly reduced: ○ KD exam time scores ($P = 0.003$) ● JVCC had significantly increased: ○ Conflict scores ($P = 0.01$)
D (27)	WM alterations	● No significance in head impact exposure between JVCC versus control between the two seasons ($P > 0.05$) ● Control group reported significantly decreased MD, AD, RD from: ○ PrY1 to PoY1 ($P < 0.05$) ○ PrY1 to PrY2 ($P < 0.05$) ○ PoY1 to PrY2 ($P < 0.05$)
E (28)	WM alterations	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● From PrY1 to PoY1, the control group reported significantly reduced: ○ MD ($P = 0.011$) ○ AD ($P = 0.021$) ○ RD ($P = 0.018$) ● From PrY1 to PoY1, the control group reported significantly increased: ○ FA ($P = 0.046$)
F (29)	Stability of functional and structural brain network activation	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● Control group reported significantly increased: ○ Modularity ($P < 0.05$) ○ rs-fMRI global clustering coefficients ($P < 0.032$) ○ DTI modularity ($P = 0.042$)
G (30)	WM alterations	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● From PrY1 to PoY1, the control group reported significantly decreased: ○ MD ($P < 0.05$) ○ AD ($P < 0.05$) ○ RD ($P < 0.05$)
H (31)	Brain activation associated with TWM	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● The control group reported a significant increase in fMRI BOLD response ($P < 0.05$) that was associated with the total number of head impacts and g-force ($P < 0.05$)
I (32)	Brain diffusivity	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● From PrY1 to PoY1, the control group reported significant changes in: ○ WM integrity ($P < 0.05$) ○ DTI change ($P < 0.05$)
J (33)	Brain activation and cognitive performance	● No significance in head blast exposure between JVCC versus control ($P > 0.05$) ● The control group reported significantly increased: ○ Post-training fMRI (2-back >0-back) brain activation ($P < 0.05$) ● The control group reported a significant correlation between brain activation and average peak pressure ($r = 0.943$, $P = 0.001$)

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Study	Outcome	Results
K (34)	Brain activity and memory performance	● No significance in head impact exposure between JVCC versus control ($P > 0.05$) ● From PrY1 to PoY1, the control group reported significantly increased: ○ Brain activation ($P < 0.05$)
L (35)	WM changes and brain network responses	● JVCC reported a significantly higher number of head impact collisions >50 g-force than the control group ($P = 0.024$) ● From PrY1 to midseason, the control group reported a significantly increased: ○ RD ($P < 0.05$) ○ MD ($P < 0.05$) ● The control group had significantly increased: ○ NBA scores($P < 0.0006$)
M (36)	Baroreflex and/or brain blood flow	● JVCC significantly increased: ○ Jugular vein cross-section area in men and women at the end of inhalation ($P = 0.003$) ● JVCC significantly reduced: ○ Carotid artery cross-sectional area at the end of exhalation ($P = 0.022$)
N (37)	Electrocortical behavior	● No significance in head blast exposure between JVCC versus control ($P > 0.05$) ● The control group had longer periods of electrocortical behavior using RQA vertical line measure after breacher training
O (38)	WM alterations	● No significance in head blast exposure between JVCC versus control ($P > 0.05$) ● JVCC reported a significant moderating effect on increased average pulse pressure ($P = 0.048$) ● From preblast to postblast exposure, the control group had significantly increased: ○ NDI ($P < 0.05$) ○ Alteration of NDI ($P < 0.05$)

IJV, internal jugular vein; rs-fMRI, resting-state functional magnetic resonance imaging; fMRI, functional magnetic resonance imaging; TWM, total working memory; RS, resting state; WM, white matter; ANT, attention network task; KD, King–Devick test; MD, mean diffusivity; AD, axial diffusivity, RD, radial diffusivity; FA, fractional anisotropy; PrY1, preseason year 1; PoY1, postseason year 1; PrY2, preseason year 2; PoY2, postseason year 2; BOLD, blood oxygen level dependent; RQA, recurrence quantification analysis; NDI, neurite density index; BNA, brain network activation on EEG; DTI, diffusion tensor imaging.

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between brain activation and average peak pulse pressure ($r = 0.943$, $P = 0.001$) (33). Control participants were found to have longer periods of electrocortical behavior using EEG measures after breacher training (37). The control group had significantly increased preblast to postblast NDI and great alteration of NDI compared to the JVCC group ($P < 0.05$) (Table 2) (38).

Certainty of Evidence

Certainty of evidence was conducted using the Cochrane Grading of Recommendations Assessment, Development, and Evaluation (GRADE) method. The outcomes of certainty of evidence are reported in Table 3.

Discussion

This analysis was conducted to determine the effect of JVCCs on athletes and adults in high-head impact scenarios. As previously stated, concussion rates are very prevalent among athletes at the high school, collegiate, and professional levels (2). The JVCC was implemented based on the slosh theory. The theory is that compression of the internal jugular vein will increase the amount of blood in the brain, which may reduce the amount of brain slosh on head impact (21). The mitigation of brain movement and the resulting sloshing effect during head impact collisions is potentially important for affected individuals. A study on former college football players reported participants with increased head impact exposure were found to have a decreased hippocampus size, which may have impacts on memory (40). Although the results are preliminary, identifying a device such as the JVCC may reduce the short- and long-term impact of repetitive head impact exposure by decreasing the sloshing effect on the brain. Table 3 outlines the certainty of evidence for JVCC versus control using the Cochrane GRADE method with outcomes rating low to moderate (39).

Physiologically, participants who wore a JVCC were found to have an increased internal jugular vein (IJV) cross-sectional area and decreased internal carotid artery cross-sectional area (Table 2) (24,36). While more research needs to be conducted on humans, current research on rats has found that increased IJV volume protects against blast exposure (41). Blast exposure, similar to head impact exposure, may lead to changes in WM and areas of brain activation (33,37,38). Peak pressure exposure was reported to have a positive correlation with increased brain activity in the control group (33), while the JVCC had a moderating effect on peak pressure (Table 2) (38). Military personnel who were exposed to high-risk blasts had an increased risk of developing a traumatic brain injury (TBI), and affected individuals also reported symptoms of headaches, tinnitus, and trouble sleeping (42,43). The JVCC has multipurpose uses and should be further looked into by those affected by head impact to reduce exposure to blasts that lead to changes in WM, concussion, or TBI.

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Table 3.
Certainty of evidence for JVCC versus control using the Cochrane GRADE Method (39).

Certainty of Evidence for JVCC versus Control				
People: High school athletics, study participants, and SWAT personal **Settings**†: Longitudinal sports season and breacher training **Intervention**†: JVCC **Comparison**†: Control				
Outcomes	Description of Outcome	Number of studies	Certainty of the evidence (GRADE)†	Comments
White matter alterations	The changes in white matter reported using measures of MD, AD, RD, and FA	7	⊕⊕⊖⊖ Low	The majority of study populations were adolescents, which is difficult to control for head impacts outside of the sports season
Head impact exposure	The exposures to the head impact between intervals	9	⊕⊕⊕⊖ Moderate	Adolescents were the primary study population with overall insignificant difference between interventions
Blast exposure	The exposure to blasts experienced by both intervals	3	⊕⊕⊕⊖ Moderate	Limited by small sample size and insignificant benefit/harm between interventions
Neurocognitive scores	Neurocognitive exam scores from participants	2	⊕⊕⊖⊖ Low	Limited by multiple per-protocol studies, lack of compliance by participants, small sample size, and population confined to adolescents
fMRI N-back working memory tasks	N-Back exam scores	3	⊕⊕⊖⊖ Low	Limited by small sample size, and one study did not have all participants complete the N-Back Working Memory Task
Concussion incidence rate	Diagnosed concussion between interventions	1	⊕⊖⊖⊖ Very low	Incidence did not significantly change between JVCC/control groups Limited by lack of compliance among participants, multiple per-protocol studies, small sample size, and population confined to adolescents

†GRADE Working Group grades of evidence.
High = This research provides a very good indication of the likely effect. The likelihood that the effect will be substantially different‡ is low.
Moderate = This research provides a good indication of the likely effect. The likelihood that the effect will be substantially different‡ is moderate.
Low = This research provides some indication of the likely effect. However, the likelihood that it will be substantially different‡ is high.
Very low = This research does not provide a reliable indication of the likely effect. The likelihood that the effect will be substantially different‡ is very high.
‡Substantially different = a large enough difference that it might affect a decision.

As seen in Table 2, changes in various WM measures, such as MD, AD, RD, and FA, were reported in the control group of football athletes (25–28,31,32), soccer athletes (26,29,30,34), hockey athletes (32), and SWAT force members (33,37,38). Postcollegiate football players with a history of concussions were found to have significant WM alterations in isotropic volume fraction (44), further expressing the risk that WM alterations may extend beyond the time of symptoms (45). Military members with a history of TBI were found to have DTI changes in different brain regions compared to controls (46). Patients with postconcussion syndrome were found to have significant changes in WM alterations compared to controls (47). While changes in WM may not initially show symptoms, the data on the long-term impact show these microchanges may persist over time. These changes may be indicative of worsening brain injury or TBI, and they may even lead to dementia (48).

Statistically significant negative changes in short-term neurocognitive outcomes were reported in control group participants, which were not seen in the JVCC groups. These participants were found to score worse on N-back tasks compared to JVCC counterparts (Table 2) (33). Similarly, a study on older adults found that a history of TBI led to a worse performance on N-back exams compared to controls (49). Athletes with postconcussion syndrome, specifically those with persistent symptoms over 6 months, were found to score lower on neuropsychology exams (47). The control group of football athletes had a worse ANT exam than the JVCC group (Table 2) (25). In another study, retired male and female athletes with a history of sports concussions were found to score lower on neurocognitive tests of memory, executive function, and motor function compared to controls (9). Decreased scores on neurocognitive exams in individuals who participate in high-impact exposure activities need to be monitored closely to mitigate the potential long-term effects of head exposure.

There were some limitations to the study. The majority of the participants in this analysis were adolescents, and the studies

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included failed to control for head impact during the athletes' off seasons. The risk of bias for WM alterations, neurocognitive scores, and N-back working memory exams was low, which demonstrates a reduced risk of bias (Fig. 2A, B). The majority of the studies were randomized at the team level, instead of at the individual level. Additionally, the JVCC is an external piece of equipment that is easily recognized by observers. Therefore, participants and observers were easily aware of who wore the JVCC, which may raise concern for targeting participants during play. However, spotters were present at every practice and at all cases of head impact data collection. Even with this in mind, almost all studies reported no significant head impact exposure difference between JVCC and control groups. As an exception, study I reported a 50 g-force difference in head impact exposure between groups (32), which was reported significant, but all other differences in head impacts between subjects in this study were considered insignificant, similar to other studies. Lastly, Dr. David Smith is the inventor of the Q-collar, which is the brand name type of JVCC. He is a medical advisor for Q30 Sports Science, LLC, which funded the majority of the studies, and may raise concern for bias (Table 1). However, our risk of bias assessment deemed the overall bias in our analysis was low (Fig. 2A, B).

The JVCC is a promising device that has been demonstrated in preliminary trials of high school athletics and SWAT personnel to limit the changes of the WM when exposed to high-impact head exposures. Whether the device can be used to reduce the risk of concussion remains unclear. Our call for future research is to assess whether the mitigation of WM may lead to a reduced risk of both concussions and long-term brain damage. Additionally, we are hopeful future research will provide a clearer conclusion of the clinical relevance and effectiveness of the JVCC. Although the literature available on the JVCC is still in its infancy, we are hopeful the JVCC will have a role in college and professional level athletes to reduce brain injury.

Conclusion

The aim of this comprehensive, up-to-date systematic review was to determine the effects of JVCCs on concussion incidence in high-impact sports and activities. Overall, the literature on JVCCs is still in its infancy. Current evidence suggests that JVCCs results in significantly reduced WM alteration, improved short-term neurocognitive outcomes, increased internal jugular vein cross-sectional area, decreased internal carotid artery cross-sectional area, and moderating effects on peak pulse pressure. The effect on JVCCs on concussion incidence remains unclear, and future research should aim to better determine the role of JVCCs in concussion prevention, brain injury, and long-term neurocognitive outcomes. The authors are hopeful that JVCCs will play a role in decreasing concussion in sports in the future.

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